

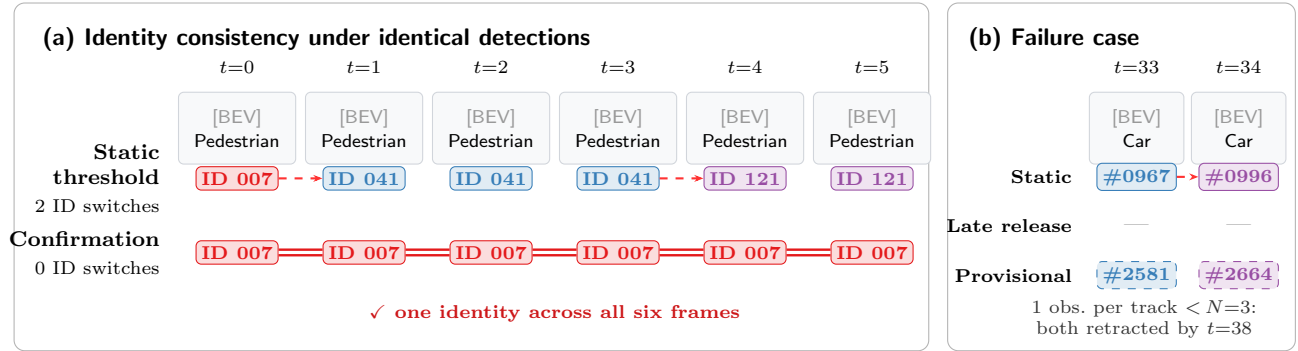


Figure 2: **Identity consistency under identical detections** (nuScenes *val*, sensor frame, detection-budget-matched arms; dashed red arrows mark identity switches; threshold identities come from the control’s own association pass). **(a)** The pedestrian of Fig. ??a (**scene-0966**). The emitted 3D box is identical in both arms in every frame — same centre, size, orientation and score (0.588–0.869, centre error ≤ 0.17 m, correctly classified throughout) — so the rows differ only in the identity attached to it. The threshold assigns three identities; confirmation with retrospective emission assigns one. **(b)** The cost: a correctly classified, well-localised car in **scene-0104**, annotated in two keyframes and held by the frozen association as two one-observation tracks, two short of $N=3$. The threshold emits it under two identities; late release never emits it; provisional emission emits both observations and retracts both tracks by $t=38$.